

Practice Karls

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1 Introduction

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2 TikZ Example

The angle α is 30° in the example ($\pi/6$ in radians). The sine of α , which is the height of the red line, is

$$\sin \alpha = 1/2. \quad (1)$$

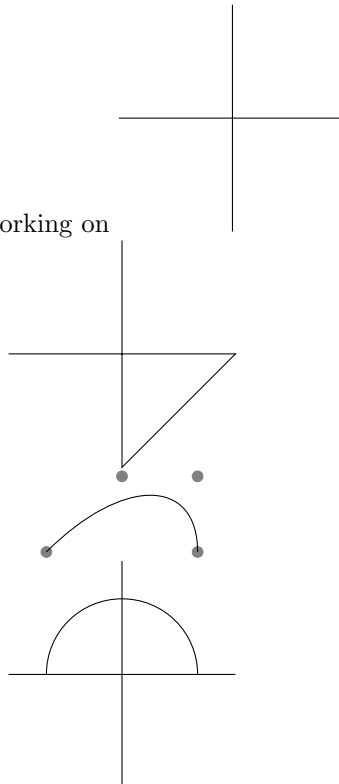
By the Theorem of Pythagoras we have $\cos^2 \alpha + \sin^2 \alpha = 1$. Thus the length of the blue line, which is the cosine of α , must be

$$\cos \alpha = \sqrt{1 - 1/4} = \frac{1}{2}\sqrt{3}. \quad (2)$$

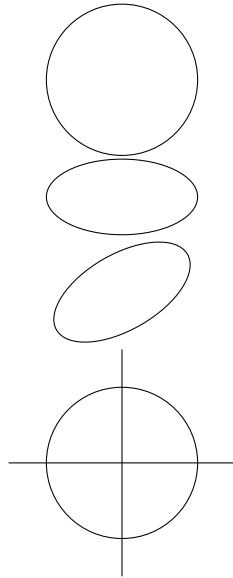
This shows that $\tan \alpha$, which is the height of the orange line, is

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{1/2}{\sqrt{3}/2} = 1/\sqrt{3}. \quad (3)$$

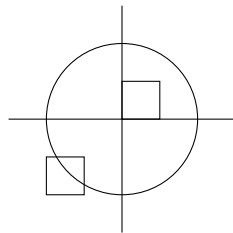
we are working on



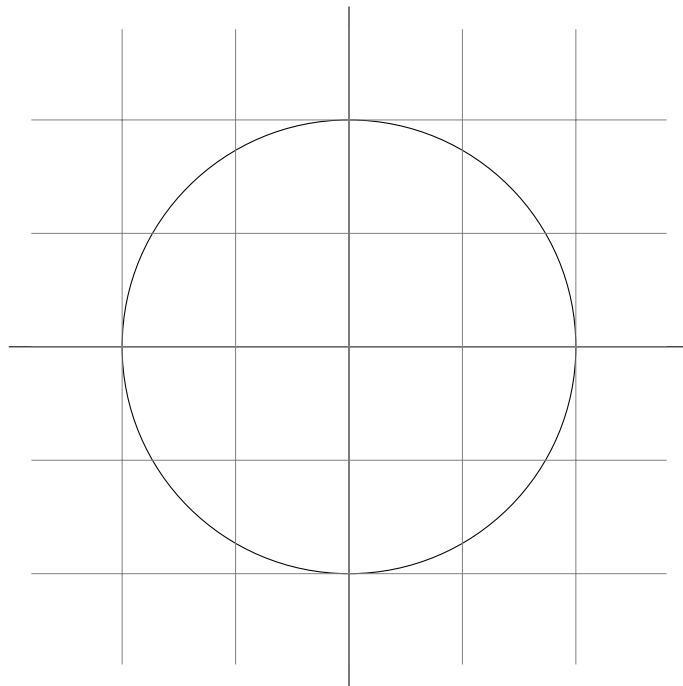
3 Circle Path Construction



4 Rectangle Path Construction



5 Grid Path Construction



6 Adding a Touch of Style

